CSE4088 Introduction to Machine Learning

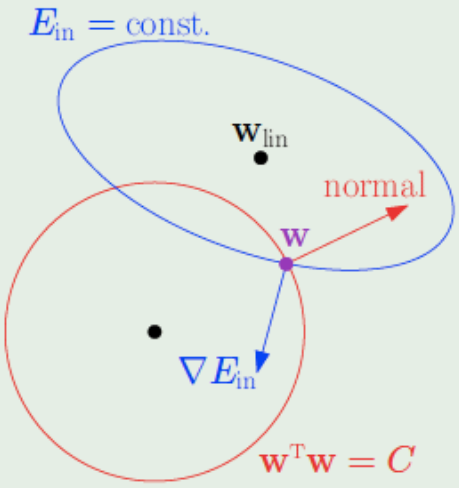
Validation

Slides are adopted from lecture notes of Yaser Abu-Mostafa

Review of last lecture

- Regularization

constrained $\longrightarrow$ unconstrained



Minimize $E_{\text{aug}}(\mathbf{w}) = E_{\text{in}}(\mathbf{w}) + \frac{\lambda}{N}\mathbf{w}^{\mathsf{T}}\mathbf{w}$

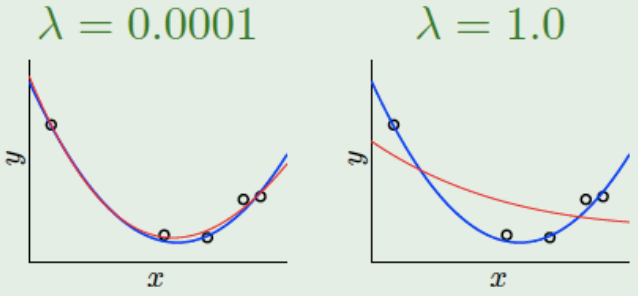
- Choosing a regularizer

$$E_{\text{aug}}(h) = E_{\text{in}}(h) + \frac{\lambda}{N}\Omega(h)$$

$\Omega(h)$: heuristic $\longrightarrow$ smooth, simple h

most used: **weight decay**

λ: principled; validation



Outline

- The validation set and the difference from test set
  - Size of the validation set is important
  - Why do we call it a «validation set» ?
- Model selection is the main task of validation.
- Cross validation
  - Trick: use all data for training, and all data for testing

Validation versus regularization

In one form or another, $E_{\text{out}}(h) = E_{\text{in}}(h) + \text{overfit penalty}$

Regularization:

$$E_{\text{out}}(h) = E_{\text{in}}(h) + \underbrace{\text{overfit penalty}}_{\text{regularization estimates this quantity}}$$

Validation:

$$\underbrace{E_{\text{out}}(h)}_{\text{validation estimates this quantity}} = E_{\text{in}}(h) + \text{overfit penalty}$$

Analyzing the estimate of E_{out}

On out-of-sample point $(\mathbf{x}, y)$, the error is $\mathbf{e}(h(\mathbf{x}), y)$

Squared error: $(h(\mathbf{x}) - y)^2$

Binary error: $[\![h(\mathbf{x}) \neq y]\!]$

$$\mathbb{E}\left[\mathbf{e}(h(\mathbf{x}), y)\right] = E_{\text{out}}(h)$$

$$\text{var}\left[\mathbf{e}(h(\mathbf{x}), y)\right] = \sigma^2$$

Problem: Using a single point for test is not reliable since its variance will be large.

From a point to a set

On a validation set $(\mathbf{x}_1, y_1), \cdots, (\mathbf{x}_K, y_K)$, the error is $E_{\text{val}}(h) = \frac{1}{K} \sum_{k=1}^{K} \mathbf{e}(h(\mathbf{x}_k), y_k)$

$$\mathbb{E}\left[E_{\text{val}}(h)\right] = \frac{1}{K} \sum_{k=1}^{K} \mathbb{E}\left[\mathbf{e}(h(\mathbf{x}_k), y_k)\right] = E_{\text{out}}(h)$$

$$\text{var}\left[E_{\text{val}}(h)\right] = \frac{1}{K^2} \sum_{k=1}^{K} \text{var}\left[\mathbf{e}(h(\mathbf{x}_k), y_k)\right] = \frac{\sigma^2}{K}$$

(since points are independent)

(variance decreased: Large K is good)

$$E_{\text{val}}(h) = E_{\text{out}}(h) \pm O\left(\frac{1}{\sqrt{K}}\right)$$

Problem: K is not free. We have N points in total, and we are going to use K of them for validation and the rest for training. Every time we take a point for validation, we take it away from training.

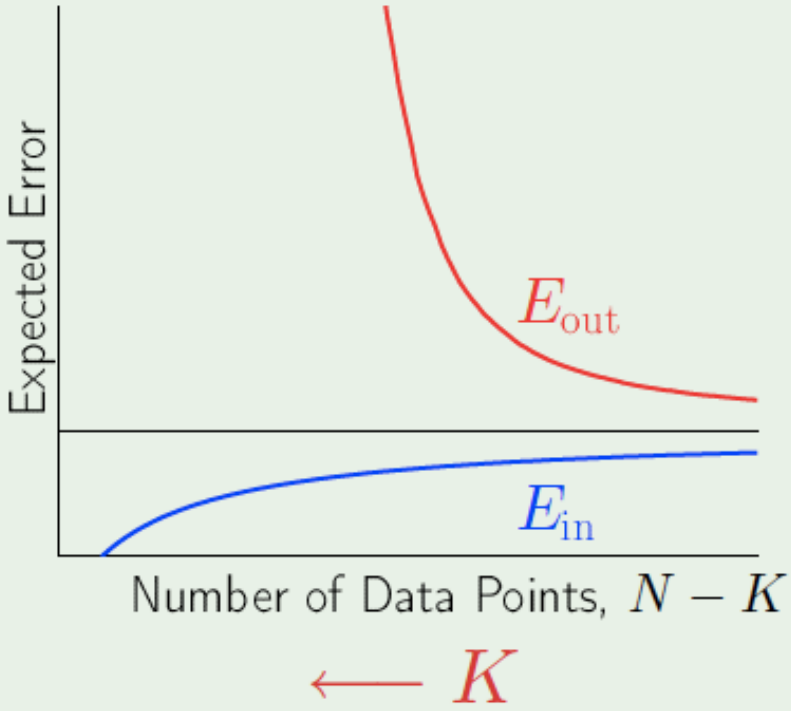
K is taken out of N

Given the data set $\mathcal{D} = (\mathbf{x}_1, y_1), \cdots, (\mathbf{x}_N, y_N)$

$\underbrace{K \text{ points}}_{\mathcal{D}_{\text{val}}} \rightarrow$ validation $\quad \underbrace{N-K \text{ points}}_{\mathcal{D}_{\text{train}}} \rightarrow$ training

$O\left(\frac{1}{\sqrt{K}}\right)$: Small $K \implies$ bad estimate

Large $K \implies$?



-- Small K will result in a bad estimate for E_{out} because its variance around the true value will be large.
-- Large K moves us to the left direction on the learnig curve, since the number of training examples decrease. Number of training points decrease. E_{out} increases.

K is put back into N

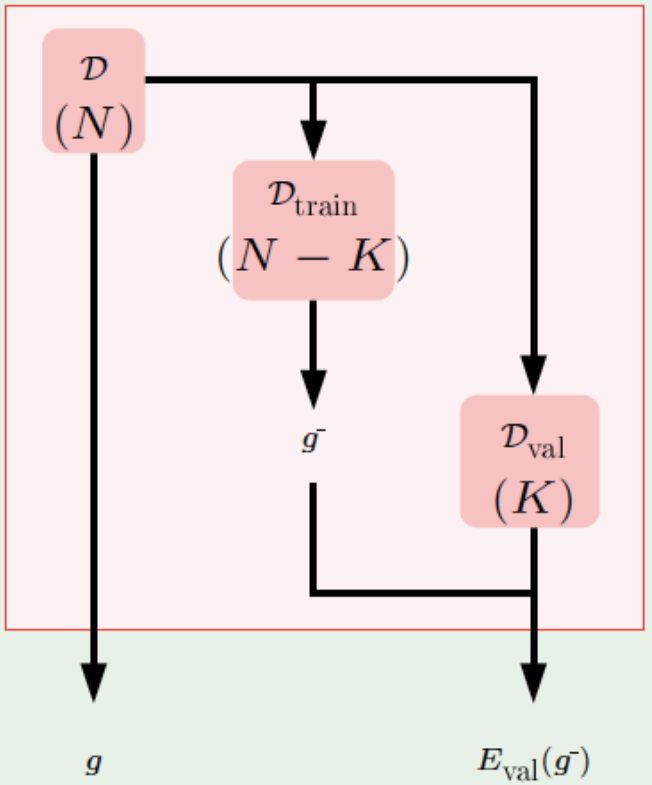
$$\begin{array}{ccccc} \mathcal{D} & \longrightarrow & \mathcal{D}_{\text{train}} & \cup & \mathcal{D}_{\text{val}} \\ \downarrow & & \downarrow & & \downarrow \\ N & & N-K & & K \end{array}$$

$$\mathcal{D} \implies g \qquad \mathcal{D}_{\text{train}} \implies g^-$$

$$E_{\text{val}} = E_{\text{val}}(g^-) \qquad \text{Large } K \implies \text{bad estimate!}$$

Rule of Thumb:

$$K = \frac{N}{5} \qquad \text{(useful in practice)}$$



After you use part of the data for validation (set your parameters etc.) you train with all of the data you have since that is your best bet of getting a good hypothesis.
We give g but report E_{val}(g-). When K is large, discrepency between g and g- is bigger.
We do not want K to be too small in order to avoid fluctuations.
We do not want K to be too large in order not to be far away from what we are reporting.

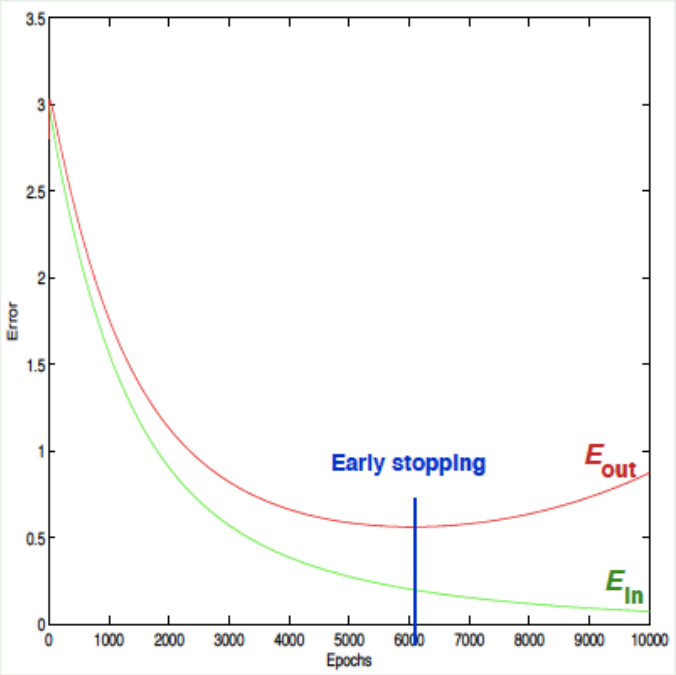
Why «validation»

$\mathcal{D}_{\text{val}}$ is used to make learning choices

If an estimate of E_{out} affects learning:

the set is no longer a **test** set!

It becomes a validation set



Validation set is **used to make choices** about the model. Hence, it plays a role in learning.

What's the difference?

Test set is unbiased; validation set has optimistic bias

Two hypotheses h_1 and h_2 with $E_{\text{out}}(h_1) = E_{\text{out}}(h_2) = 0.5$

Error estimates $\mathbf{e}_1$ and $\mathbf{e}_2$ uniform on $[0, 1]$

Pick $h \in \{h_1, h_2\}$ with $\mathbf{e} = \min(\mathbf{e}_1, \mathbf{e}_2)$

$\mathbb{E}(\mathbf{e}) < 0.5$ optimistic bias

(Denote e < 0.5 by 1:

1 1
1 0
0 1
0 0

Three of the four possibilies give us $\min(e_1,e_2) < 0.5$. Then the expected value of e is less than 0.5 since the probability that minimum will be less than 0.5 is 75%.)

For a proof see:
https://danieltakeshi.github.io/2016/09/25/the-expectation-of-the-minimum-of-iid-uniform-random-variables/

Outline

- The validation set
- Model selection
- Cross validation

Using $\mathcal{D}_{\text{val}}$ more than once

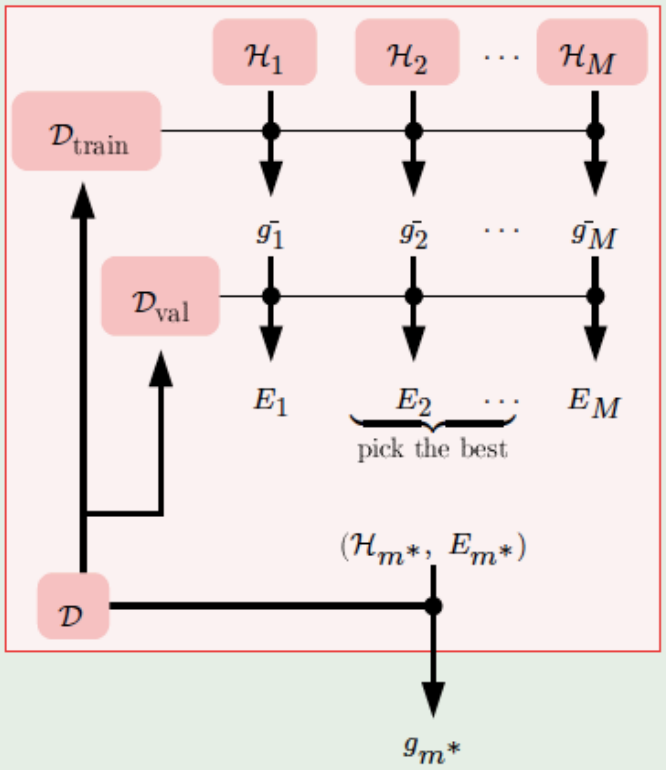
M models $\mathcal{H}_1, \ldots, \mathcal{H}_M$

Use $\mathcal{D}_{\text{train}}$ to learn g_m^- for each model

Evaluate g_m^- using $\mathcal{D}_{\text{val}}$:

$$E_m = E_{\text{val}}(g_m^-); \quad m = 1, \ldots, M$$

Pick model $m = m^*$ with smallest E_m



We want to model selection in a principled way using the validation set. Model selection examples:
1) linear models or neural networks or SVM?
2) We could be using only polynomial models and trying to select the order of the polynomial (2nd , 5th etc.)
3) We could be using a 5th order polynomial and trying to estimate the best lambda of the regularization term. (0.01, 0.1, 1 etc.)

The bias

We selected the model $\mathcal{H}_{m^*}$ using $\mathcal{D}_{\text{val}}$

$E_{\text{val}}(g^-_{m^*})$ is a biased estimate of $E_{\text{out}}(g^-_{m^*})$

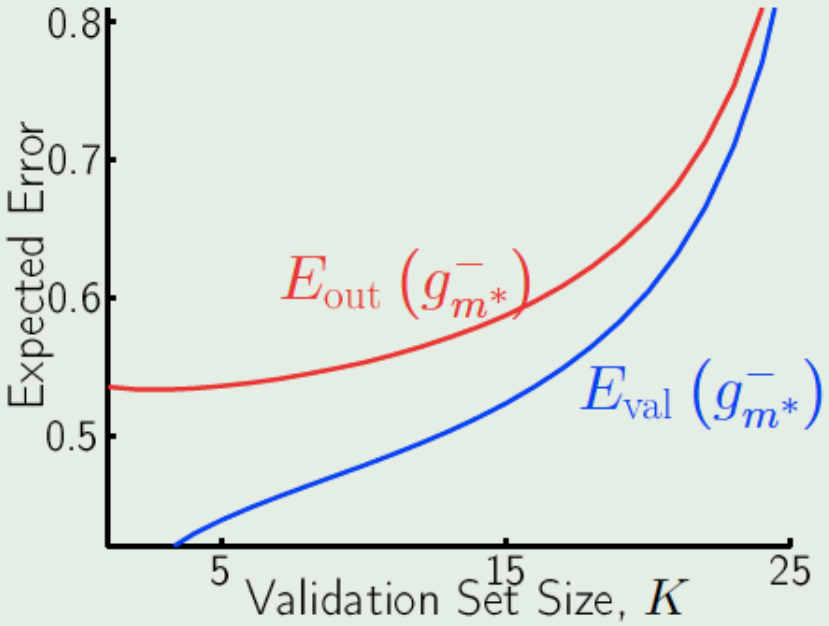
Illustration: selecting between 2 models

Why are the curves going up?

Since as K increases, we have less training data, which gives a worse model.

Why are they becoming closer to each other?

When we have bigger K, E_{val} estimates E_{out} better (more reliable).



Choose between two models: second order polynomials and fifth order polynomials. We repeat the experiment many times. In each run we pick the hypothesis with the smaller E_{val}. The number of samples in the data set is 30 and we use K of them for validation.

How much bias

For M models: $\mathcal{H}_1, \ldots, \mathcal{H}_M$ $\mathcal{D}_{\text{val}}$ is used for "training" on the **finalists model**:

$$\mathcal{H}_{\text{val}} = \{g_1^-, g_2^-, \ldots, g_M^-\}$$

Back to Hoeffding and VC!

$$E_{\text{out}}(g_{m*}^-) \leq E_{\text{val}}(g_{m*}^-) + O\left(\sqrt{\frac{\ln M}{K}}\right)$$

regularization λ early-stopping T

Rule of thumb: If your validation set consists of 100 samples, you can choose 2-3 parameters. If you choose 20 parameters, you are contaiminating. Estimate will be ruined.

Data contamination

Error estimates: E_{in}, E_{test}, E_{val}

Contamination: Optimistic (deceptive) bias in estimating E_{out}

Training set: totally contaminated

Validation set: slightly contaminated

Test set: totally 'clean'

You can not rely on E_{in} to estimate E_{out}

If you are using data for training (or to make choices), you are contaminating its ability to estimate the real performance E_{out}.

Outline

- The validation set
- Model selection
- Cross validation

The dilemma about K

The following chain of reasoning:

(Want to know this) (Can calculate this)

$$E_{\text{out}}(g) \approx E_{\text{out}}(g^-) \approx E_{\text{val}}(g^-)$$

(small K) (large K)

highlights the dilemma in selecting K:

Can we have K both small and large? ☺

Leave one out cross validation

$N-1$ points for training, and 1 point for validation!

$$\mathcal{D}_n = (\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_{n-1}, y_{n-1}), \cancel{(\mathbf{x}_n, y_n)}, (\mathbf{x}_{n+1}, y_{n+1}), \ldots, (\mathbf{x}_N, y_N)$$

Final hypothesis learned from $\mathcal{D}_n$ is g_n^-

$$\mathbf{e}_n = E_{\text{val}}(g_n^-) = \mathbf{e}\left(g_n^-(\mathbf{x}_n), y_n\right)$$

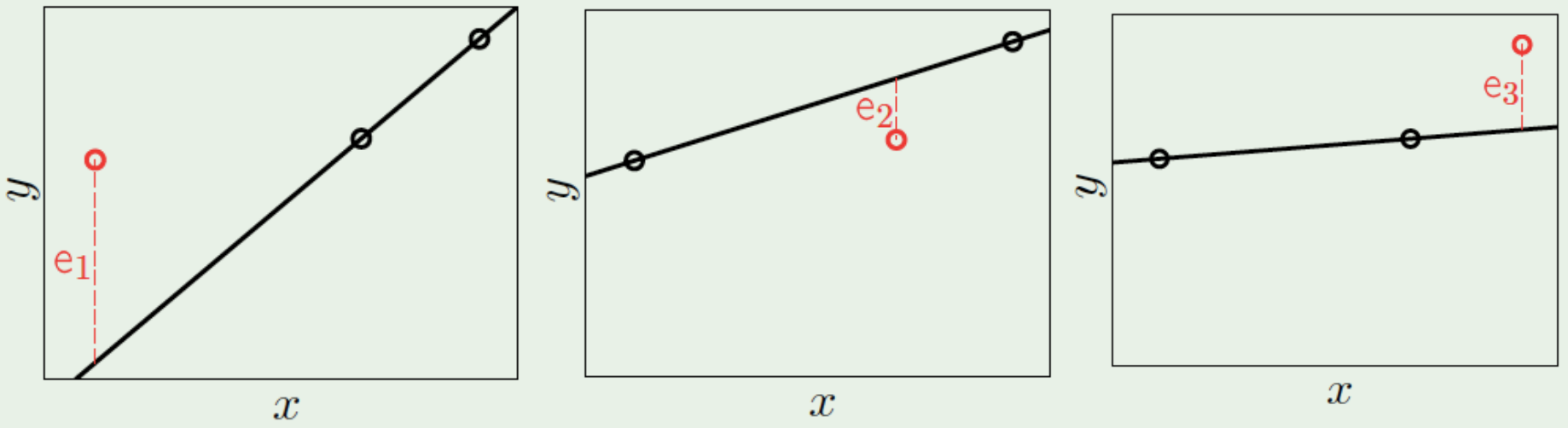
cross validation error: $E_{\text{cv}} = \frac{1}{N}\sum_{n=1}^{N} \mathbf{e}_n$

Repeat the same for each sample.

It is as if we are validating using N points and training on N-1 points.

Although e_n are not independent, works well in practice.

Illustration of cross validation

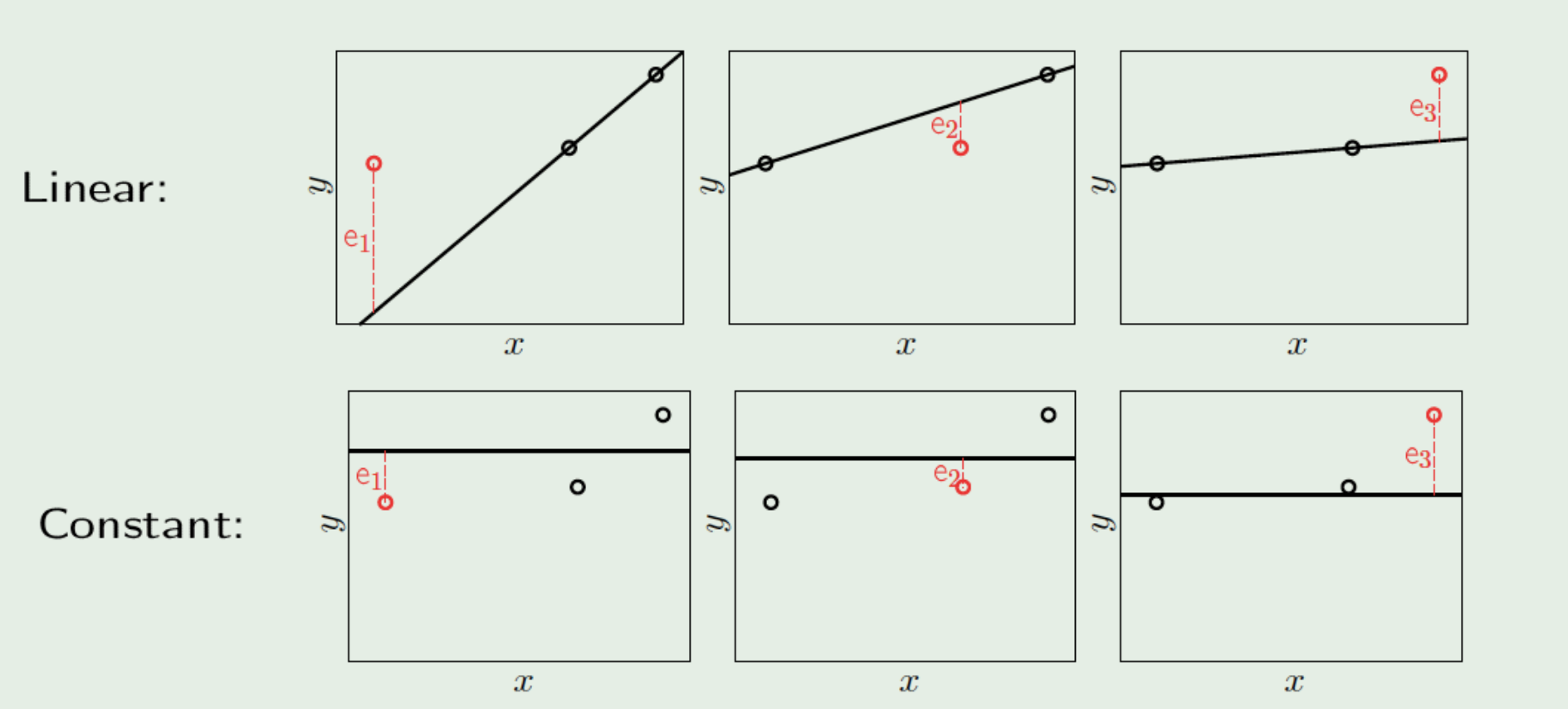


$$E_{\text{cv}} = \frac{1}{3}\left(e_1 + e_2 + e_3 \right)$$

(Cross validation error:
Estimate of E_{out} for the linear model.)

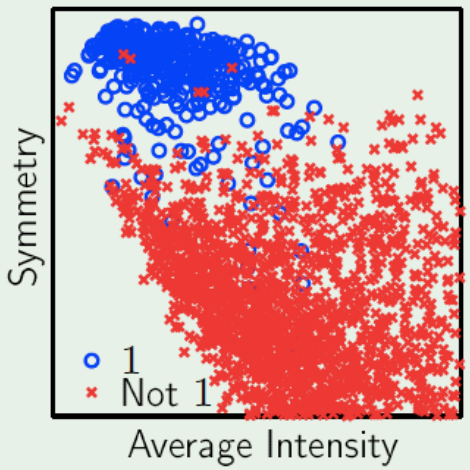
Goal: Given 3 data points, want to fit a curve.
Estimate E_{out} for a linear model using cross validation.

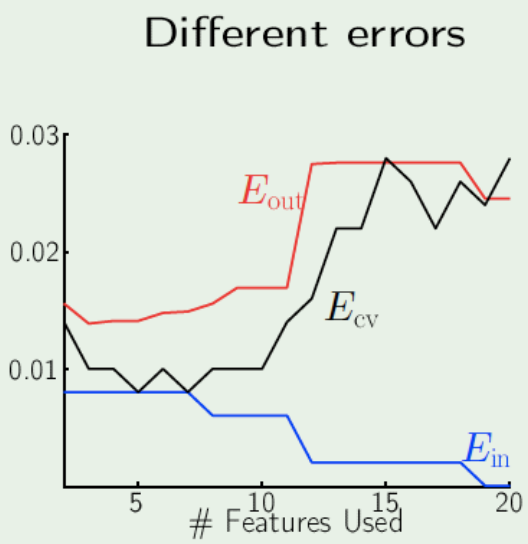
Model selection using CV



Constant model wins. The data was generated from a constant model with noise.

Cross validation in action



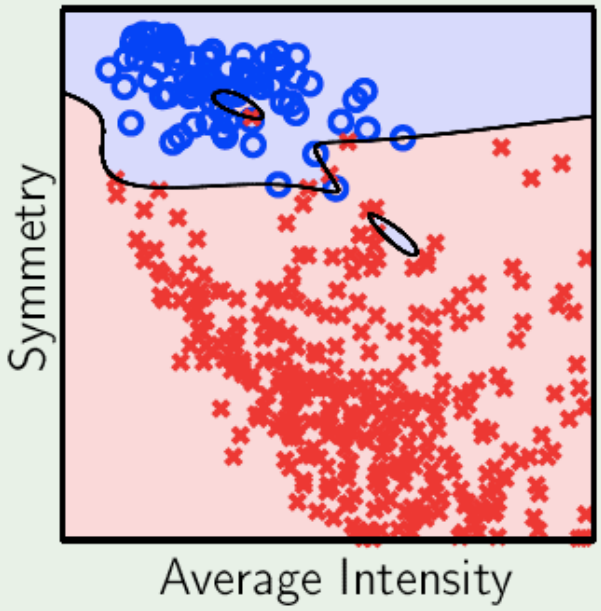


$$(1, x_1, x_2) \rightarrow (1, x_1, x_2, x_1^2, x_1x_2, x_2^2, x_1^3, x_1^2x_2, \ldots, x_1^5, x_1^4x_2, x_1^3x_2^2, x_1^2x_2^3, x_1x_2^4, x_2^5)$$

If we were to pick a model based on E_{in} we would use all the 20 dimensions. E_{CV} is minimized between 5-7 dimensions; we can take 6 feature dimensions as the model selected by cross validation.

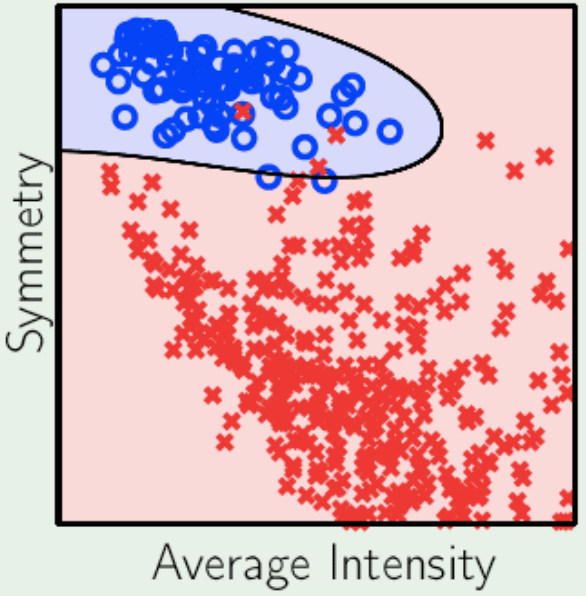
The result

without validation



$E_{\text{in}} = 0\%$ $E_{\text{out}} = 2.5\%$

with validation

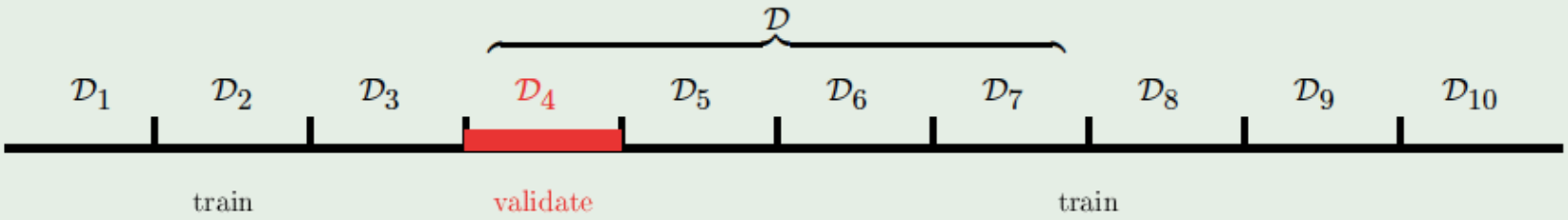


$E_{\text{in}} = 0.8\%$ $E_{\text{out}} = 1.5\%$

Leave more than one out

Leave one out: N training sessions on $N-1$ points each

More points for validation?



$\frac{N}{K}$ training sessions on $N-K$ points each

10-fold cross validation: $K = \frac{N}{10}$

Using Cross Validation

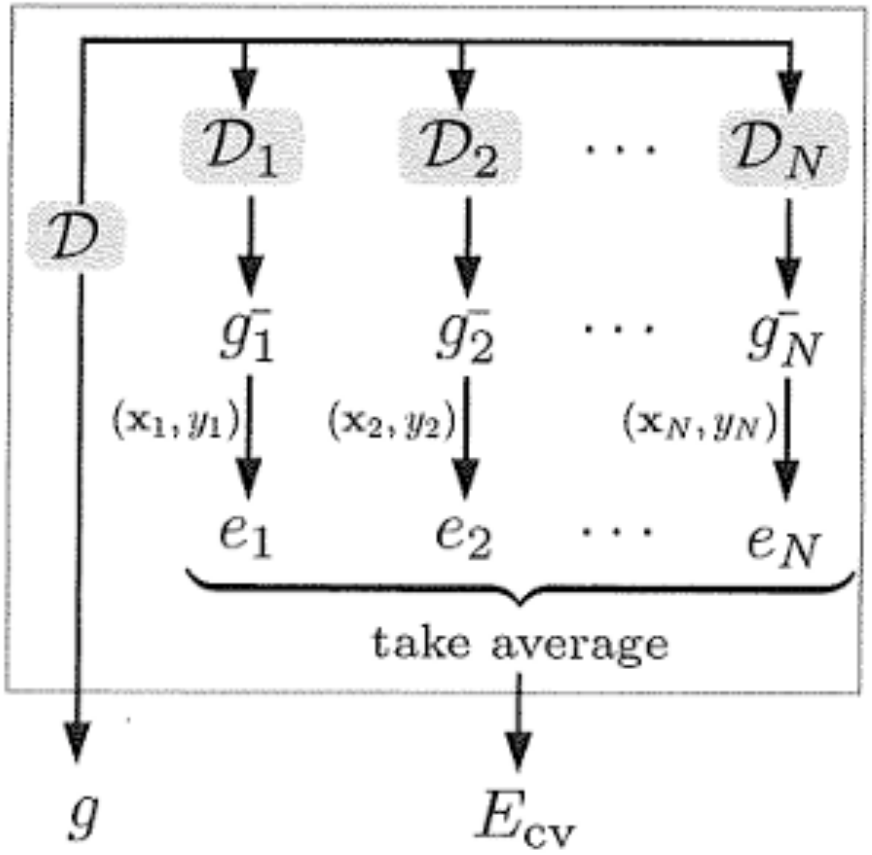


Figure 4.14: Using cross validation to estimate E_{out}

Cross Validation for Selecting the Regularization Parameter

Cross validation for selecting λ:

1: Define M models by choosing different values for λ in the augmented error: $(\mathcal{H}, \lambda_1), (\mathcal{H}, \lambda_2), \ldots, (\mathcal{H}, \lambda_M)$
2: **for** each model $m = 1, \ldots, M$ **do**
3: Use the cross validation module in Figure 4.14 to estimate $E_{\text{cv}}(m)$, the cross validation error for model m.
4: Select the model m^* with minimum $E_{\text{cv}}(m^*)$.
5: Use model $(\mathcal{H}, \lambda_{m^*})$ and all the data $\mathcal{D}$ to obtain the final hypothesis g_{m^*}. Effectively, you have estimated the optimal λ.